Abstract

Billions of musical scores remain locked in physical scans, fundamentally inaccessible to computational analysis, limiting digital musicology and cultural preservation efforts [1, 2]. Automating transcription of these documents through Optical Music Recognition (OMR) presents significant challenges in representation learning. Due to the absence of large-scale annotated datasets of real-world scans, current models must depend entirely on zero-shot simulation-to-reality transfer [2]. Existing end-to-end approaches often resort to brittle heuristic pipelines or excessively large models to stabilize transcription [2–4].

We introduce ***Transcoda****, a compact 59M-parameter vision-encoder-decoder framework trained exclusively on synthetic data to study zero-shot OMR. We show that emphasizing target canonicalization over model size enables stable training and convergence in only 5 hours on a single consumer-grade GPU. Transcoda combines a deterministic data pipeline that removes structural ambiguity from training targets with an optional stateful constrained decoding engine. Unconstrained beam search yields the best raw sequence-matching accuracy, while constrained decoding provides a strict formal validity guarantee for downstream rendering.*

Transcoda achieves the best score among compared systems on our standardized Verovio benchmark at 18.22% OMR-NED. On the challenging real-world Polish dataset, our compact model lowers error to 76.03% OMR-NED and consistently outperforms baseline methods that are an order of magnitude larger. This result does not solve real-world OMR. Instead, it quantifies the severity of the sim-to-real gap. Our ablations show that target canonicalization is the main source of generation stability, while heuristic synthetic augmentations do not close transfer to historical scans yet.

Code, model weights, and datasets are available at `https://github.com/btrkeks/transcoda`.

1. Introduction

Optical Music Recognition as Structured Sequence Modeling. Optical Music Recognition (OMR) aims to translate images of sheet music into machine-readable symbolic formats. While modern vision-language models have achieved remarkable progress in optical character recognition [5, 6], OMR remains fundamentally more challenging. Unlike natural language, music notation encodes a dense two-dimensional relational structure: symbols overlap, share spatial anchors (e.g., staff lines), and interact through global rhythmic constraints. As a result, OMR is not merely visual recognition but a structured sequence modeling problem with strict syntactic and semantic dependencies [1, 2].

The Zero-Shot Sim-to-Real Regime. A defining constraint of modern OMR is the absence of large-scale annotated real-world datasets. Consequently, state-of-the-art systems are trained entirely on synthetic renderings and evaluated zero-shot on physical scans [2]. While recent end-to-end image-to-sequence models have significantly improved transcription quality [3,4], they exhibit severe degradation under real-world distribution shifts. In particular, small visual deviations, such as noise, occlusions, or engraving irregularities, often trigger catastrophic sequence collapse: the autoregressive decoder produces structurally invalid outputs, violating global musical constraints and rendering predictions unusable. In addition to visual noise, real-world scores exhibit substantially higher layout density and sequence length than typical synthetic training data. This induces a secondary distribution shift in the autoregressive decoding regime, where models must extrapolate from short, sparse sequences to dense, long-range structured outputs.

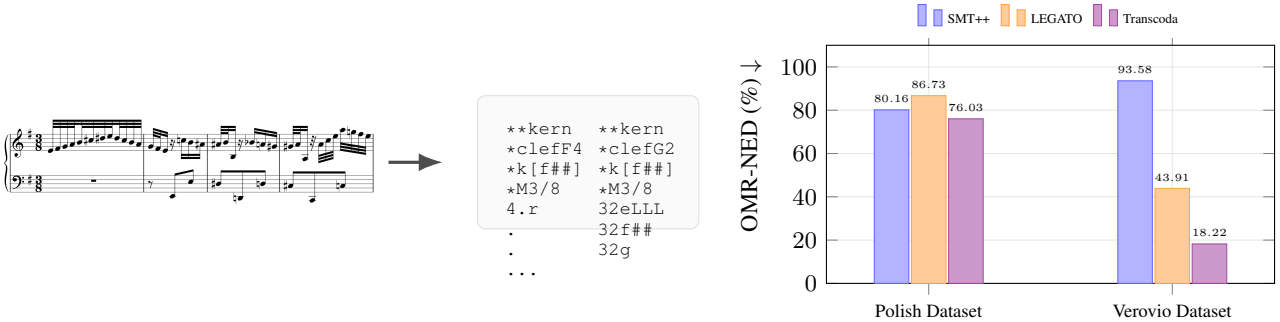


Figure 1. Left: Transcoda maps a raw score image to an aligned `**kern` transcription. Right: the same compact 59M-parameter model reaches the lowest OMR-NED among the compared systems on both the synthetic Verovio benchmark and the real-world Polish Dataset.

Failure Mode: Target Entropy and Structural Instability. We argue that this failure is not primarily due to insufficient model capacity, but rather to *uncontrolled target entropy*. In existing pipelines, the same visual score can map to many syntactically different yet semantically equivalent sequences due to flexible encoding conventions (e.g., in `**kern`). This one-to-many mapping forces the decoder to model annotator variability in addition to the underlying musical structure, amplifying uncertainty during training. Under distribution shift, this ambiguity compounds, leading to unstable generation and sequence collapse.

A Data-Centric Perspective on OMR. In this work, we investigate whether reducing target entropy, rather than increasing model scale, can stabilize zero-shot OMR. We introduce **Transcoda**, a compact 59M-parameter vision-encoder-decoder framework trained exclusively on synthetic data. Our approach is built on a simple principle: if the target distribution is deterministic and structurally aligned with the decoding grammar, autoregressive sequence modeling becomes stable even under severe visual shift.

To this end, we construct a data engine that enforces a one-to-one mapping between visual inputs and symbolic targets through strict canonicalization, while simultaneously increasing visual diversity via domain-aware augmentations. At inference time, we optionally apply a stateful constrained decoding engine that enforces global syntactic and rhythmic consistency when a downstream application requires formally valid output.

Overview of the Approach. Transcoda combines three components: (1) a compact ConvNeXt-V2 + Transformer architecture optimized for full-page transcription, (2) a deterministic data pipeline that removes structural ambiguity from training targets, and (3) an optional constrained decoding engine that enforces the formal grammar of music notation during generation.

Our experiments show that the second component is the main stabilizer. Canonicalization removes the one-to-many mapping between image and target, allowing the autoregressive decoder to learn a cleaner visual-semantic alignment. The constrained decoder serves a different role: it guarantees formal renderability for downstream parsers, but can trade raw edit distance for validity.

Contributions. Our contributions are as follows:

1. We identify *target entropy* as a primary bottleneck in zero-shot OMR and show that reducing structural ambiguity in the training data significantly stabilizes autoregressive sequence generation.

2. We introduce **Transcoda**, a compact (59M parameter) vision-encoder-decoder architecture that achieves the best score among compared systems on our synthetic benchmark and outperforms much larger baselines on real-world scans.

3. We propose a data-centric training pipeline combining strict symbolic canonicalization, asymmetric semantic augmentation, and domain-aware visual degradation.

4. We develop a stateful, layered constrained decoding engine that strictly enforces 2D musical grammar and rhythmic mathematics during inference. This provides a critical mechanism to guarantee 100% formally valid and renderable transcriptions for downstream applications.

5. We release a standardized Verovio-based synthetic benchmark to isolate notation modeling performance from visual domain shift, providing a clean evaluation protocol for future OMR systems.

2. Related Work

Traditional OMR Methods. Initial Optical Music Recognition (OMR) relied on rule-based image processing and template matching, performing well on clean, standardized scores but failing under realistic conditions such as layout variations or degradation [7, 8].

Deep Learning Approaches. Modern OMR transitioned to deep learning, initially with hybrid systems combining learned representations with heuristic grammars or modular pipelines, still vulnerable to cascading errors [9, 10]. Recent end-to-end approaches further improved transcription quality by modeling full-page context directly from images [4, 11–13]. Nonetheless, these methods struggle significantly when transferring to real-world scans due to the lack of annotated datasets [2].

Synthetic-First OMR. Recent synthetic-first strategies, notably Legato [2], mitigate data scarcity by pretraining on synthetic renders. However, Legato's reliance on model scale rather than data rigor results in structural instability under real-world noise and domain shifts [2]. In contrast, we address these limitations through data rigor, using a compact model to test which design choices actually stabilize generation and which ones fail under severe domain shift.

3. Method

Transcoda is a vision-encoder-decoder model. Given a fixed-size score image, it generates a `**kern` token sequence autoregressively. Every design choice in this section targets the exact same goal: narrow the domain gap without requiring real annotated scans.

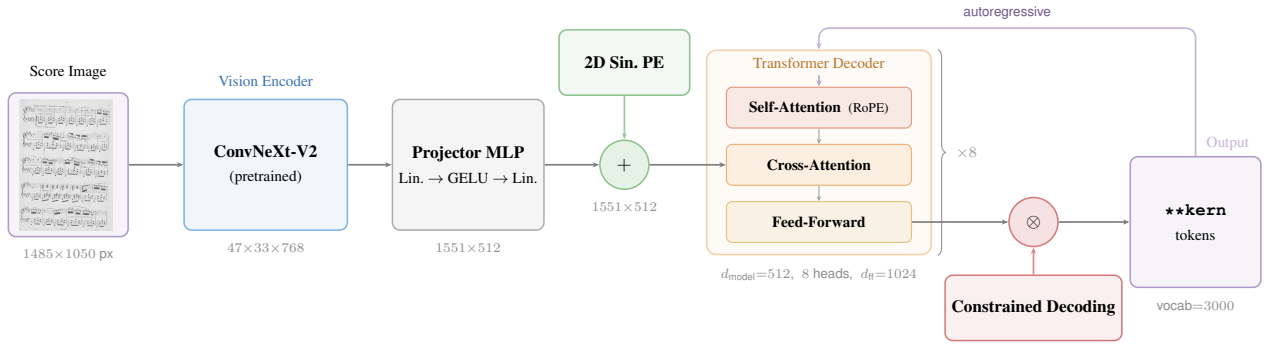


Figure 2. Transcoda architecture and inference pipeline. A pretrained ConvNeXt-V2 encoder extracts dense visual features from the raw score image. A two-layer MLP projects these features to the decoder's hidden dimension, followed by the addition of 2D sinusoidal positional encodings to retain spatial topology. An 8-layer Pre-LN Transformer decoder autoregressively generates the `**kern` sequence. During inference, a stateful constraint engine dynamically masks the output logits. This enforces formal grammar, parallel spine integrity, and rhythmic mathematics, ensuring structurally valid transcriptions even under severe visual domain shift.

3.1. Vision-Encoder-Decoder Architecture

Transcoda builds upon the standard vision-encoder-decoder setup used in prior work [13]. We deliberately retain a compact sequence-to-sequence architecture because our preliminary analyses indicated that model capacity is not the primary bottleneck in zero-shot OMR. Instead, transcription failures are driven by target entropy and the simulation-to-reality visual domain shift. By avoiding unnecessary architectural complexity, we use the model as a controlled testbed for these data-centric bottlenecks without massive parameter scaling.

Relative to prior baselines such as SMT++, we make four targeted architectural changes to maximize representation stability and throughput:

Pretrained Vision Backbone. We replace the from-scratch visual encoder with a pretrained `facebook/convnextv2-tiny-22k-224` backbone [14]. We fix input resolution at 1485×1050 pixels, maximizing throughput via static tensor shapes optimized by `torch.compile`.

Visual Projection. We insert a two-layer projection MLP with GELU activations between the encoder and decoder. This maps the downsampled visual grid to the hidden dimension of the decoder.

Decoder Design. We use an 8-layer Pre-LN Transformer decoder with GELU feed-forward blocks to stabilize mixed-precision training. The decoder uses $d_{model} = 512$, $d_{ff} = 1024$, 8 attention heads, and RoPE self-attention.

Spatial Context. Before flattening the visual grid into a sequence, we add 2D sinusoidal positional encodings so the decoder receives explicit spatial cues [15, 16].

Finally, we cap the maximum sequence length at 2048 tokens. This threshold acts as a strict model-data contract: it bounds training memory and padding overhead while remaining long enough to capture the density of a full musical page. The final model contains approximately 59 million parameters.

3.2. Split-Space Tokenization

Formulating OMR as sequence generation requires a target text format that captures both semantic and spatial structure. We use the Humdrum `**kern` format [17]. Unlike performance-oriented formats such as MIDI or verbose markup formats such as MusicXML, `**kern` represents music as a dense two-dimensional matrix serialized into a single text stream [17].

`**kern` encodes time row by row. Concurrent musical parts are arranged horizontally and separated by tabs. A standard token encodes pitch and duration together, such as `4C` for a quarter-note middle C. Simultaneous notes within one voice are separated by spaces [17]. We train a custom BPE tokenizer over this representation with a vocabulary size of 3000, split-space merge constraints, and standard special tokens, following the standard byte-pair encoding setup [18]. Figure 3 shows a score excerpt beside the corresponding `**kern` transcription.

The decoder therefore treats `**kern` as a dense text matrix serialized into a 1D target stream. To predict this representation reliably, it must learn four linked encoding rules.

Grid structure and alignment. Time flows from top to bottom. Parallel parts are arranged in tab-delimited columns, typically one per staff. Since different staves often hold notes of different durations, the format maintains strict temporal alignment with the null token `.`. This token acts as structural padding so that every row remains synchronized at each fractional time step [17].

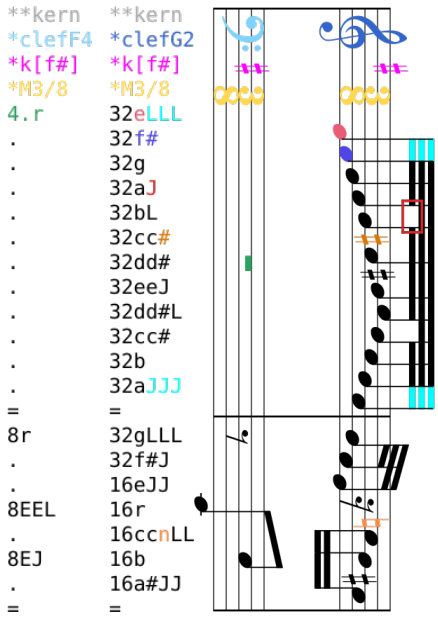


Figure 3. **The Humdrum `**kern` representation.** The 2D musical graph is linearized into a synchronous, top-to-bottom sequence. Columns encode parallel staves, while rows encode discrete time steps. Null tokens (`.`) act as padding to maintain strict temporal alignment across the matrix. Tokens compactly fuse duration, pitch, and structural metadata, for example `32eLLL` for a 1/32 note E that opens three beams [17].

Pitch and register. Pitch is written with note letters, while register is encoded through case and repetition. Lowercase letters denote higher octaves and uppercase letters lower octaves. Repeating the letter shifts the register further. Accidentals are appended as suffixes, such as `#` for sharps or `n` for naturals [17].

Duration and rhythm. Rhythmic values are encoded by the reciprocal of the note length. For example, `32` denotes a 1/32 note. Dots extend the duration, such as `4.` for a dotted quarter. Rests are marked with the `r` symbol [17].

Structural metadata. Global score state and visual grouping are embedded as explicit control tokens. These include clefs such as `*clefG2` and `*clefF4`, key signatures such as `*k[f#]`, and time signatures such as `*M3/8`. Beams are encoded with trailing markers: `L` opens a beam and `J` closes it. Repetition indicates beam depth, so `LLL` opens a tertiary beam. Barlines are marked with tokens beginning with = [17].

The essential modification is split-space behavior. In `**kern`, spaces separate simultaneous notes [17]. Standard BPE algorithms often merge across spaces and create brittle chord tokens [18]. Data-driven tokenization should compress frequent musical patterns into reusable units, such as common chords or rhythmic groups. Standard BPE does this without respecting musical syntax [18]. It can learn `4C_4E` as one token and then fail on an unseen chord variant. Implementing a split-space constraint prevents cross-space merges and maintains compositional chords at inference time. This lets the tokenizer capture valid vertical musical concepts without memorizing fixed chord surfaces that do not transfer to unseen distributions.

3.3. Data Engine for Controlled Synthetic Training

Data preparation follows a strict pipeline consisting of extraction from open digital repositories, conversion to `**kern`, and filtering of invalid files. We then apply grammar-based normalization, split-space BPE training, rendering with augmentation, and finally a post-audit on sequence length. We treat this pipeline as a data engine with three explicit stages: target canonicalization, asymmetric semantic augmentation, and domain-aware visual degradation.

Table 1. Dataset attrition through the symbolic preprocessing pipeline. All raw corpora undergo identical conversion, filtering, and normalization stages to guarantee structural consistency across distributions. The final normalized pools constitute the training and internal validation sets.

Source	Raw	Conv.	Filt.	Norm.
PDMX	234,539	234,379	176,838	176,838
Grandstaff	41,598	41,598	40,859	40,859
MuseTrainer	69	69	49	49
OpenScore Lieder	1,389	1,389	1,017	906
OpenScore Quartets	122	121	41	38
Total	**277,717**	**277,556**	**218,804**	**218,690**

The engine favors target quality over raw dataset scale. Tab. 1 details the data attrition across extraction, conversion, filtering, and normalization. Normalization maps diverse, loosely formatted transcriptions into a strict canonical structure. The attrition in this final phase occurs because a small subset of files contains complex structural edge cases. We discard these long-tail anomalies, as engineering custom formatting rules for them yields diminishing returns for dataset scale while risking target consistency. To maximize the utility of the rigorously cleaned files, we render subsets of the normalized pool multiple times under different combinations of semantic injection and visual degradation. This expands the base symbolic files into a final synthetic training set of 361,938 visually diverse images.

A critical early bottleneck in this pipeline was the brittleness of existing format converters. We observed that the standard `musicxml2hum` converter fatally crashed on over 10% of the raw PDMX corpus due to unhandled edge cases and parsing bugs in real-world MusicXML files. To prevent massive data attrition before the filtering stage, we

implemented extensive structural fixes to the converter's source code, successfully reducing the conversion failure rate to under 0.07% (as reflected in the *Conv.* column of Tab. 1). We release this hardened converter alongside our main codebase to support future large-scale dataset curation.

3.3.1 Deterministic Data Filtering.

Before canonicalization, we apply a strict, fail-fast filtering pipeline to remove pathological files and unsupported notation from the converted pool. We group these filters into three criteria:

1. *Structural Integrity:* We discard files with broken UTF-8 encoding, missing Humdrum spine terminators (`*-`), or missing initial clef declarations.
2. *Conversion Artifacts:* We apply bounding heuristics to reject severe XML-to-Humdrum conversion errors, specifically discarding files with mensural rational durations (e.g., `2%5`), impossible accidental runs (three or more), or corrupted octave spellings (seven or more repetitions). We identified these specific edge cases during early grammar validation; manual inspection confirmed they almost exclusively stem from malformed source MusicXML rather than legitimate notation.
3. *Rhythmic Validation:* Finally, files pass through an external rhythm-checking engine. Transcriptions with fundamentally broken measure mathematics are discarded, though we explicitly allow valid musical exceptions such as anacrusis (pickup measures) and incomplete final measures.

This stage deterministically prunes approximately 21% of the raw converted corpus. While aggressive, this ensures the model's capacity is devoted strictly to learning valid, parsable musical graphs rather than memorizing corrupted layouts or unrecoverable transcription errors.

3.3.2 Target Canonicalization

To eliminate the target entropy described in Sec. 1, we enforce a deterministic mapping between visual input and symbolic target by processing all training data through a strict 21-pass canonicalization pipeline. While we employ structure-aware metrics (TEDn, OMR-NED) to mitigate evaluation brittleness caused by textual divergence [19, 20], strict canonicalization remains essential to establish an unambiguous ground truth for sequence-level comparison. Early struggles on Polish scanned scores revealed that raw label noise severely impacted cross-domain performance.

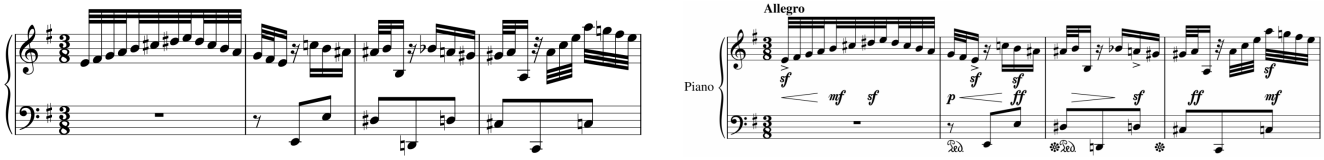
Deterministic Normalization Pipeline. The pipeline operates in strict sequence to rewrite arbitrary transcriptions into a single canonical form strictly bounded by our decoding grammar:

- **Removing Extraneous Spines:** We completely strip non-notation parallel spines, such as lyrics (`**text`) and dynamics (`**dynam`). Removing them from the ground truth ensures the model only has to predict the core musical notes. Later, during rendering, we procedurally re-inject these elements into the image as visual distractors, giving us total control over the visual noise.
- **Visual-Semantic Alignment (Passes 3–9):** The model must not be penalized for failing to predict invisible structural markup. We systematically remove non-visual elements such as playback-only grace rests used for cross-staff alignment, redundant mid-piece stria declarations, and terminal string markers (`*-`). This ensures sequence length and content correlate directly and exclusively with the rendered notation.
- **Syntactic Token Canonicalization (Passes 12–17):** A single `**kern` token encodes up to 13 properties (duration, pitch, accidentals, articulations, slurs, beams, etc.), which different sources emit in arbitrary orders. We enforce a fixed, character-level sorting hierarchy for every token component. For example, the structurally identical but textually divergent tokens `[8fJ` and `8f[J` deterministically normalize to `8f[J`. Furthermore, simultaneous notes within chords are explicitly sorted in ascending pitch order.
- **Structural Repair and Resolution (Passes 18–21):** We actively resolve contradictory annotations generated by flawed upstream MusicXML converters. For instance, we collapse conflicting accidentals (e.g., a sharp and a natural on the same token) into single valid modifiers, cap recursive slur bounds to a maximum nesting depth of two, and eliminate self-canceling tie/slur pairs. Finally, files undergo strict structural validation; any sequence with unsupported spine manipulations is discarded entirely.

Table 2. Examples of deterministic target canonicalization. Raw sequences with identical visual meanings are collapsed into a single grammatical form.

Artifact Type	Raw Input	Canonical Output	Pipeline Action
Syntactic Ambiguity	`[8fJ` *or* `8fJ[`	`8f[J`	Token syntax sorting
Contradictory XML	`4c#n`	`4cn`	Resolve conflicting accidentals
Redundant Metadata	`*met(c)` *then* `*M4/4`	`*M4/4`	Remove equivalent time signatures
Self-Canceling	`4G[]`	`4G`	Remove zero-length ties

Grammar Grounding. By removing superficial sequence variation and standardizing edge cases, this pipeline guarantees every training target belongs to the exact



(a) Base render from the canonicalized target sequence.

(b) The same sample after semantic injection. Dynamics, pedal markings, and tempo text increase visual density, but the target sequence is unchanged.

Figure 4. Asymmetric semantic augmentation. We inject musically valid performance markings into the rendered score while preserving the same canonical `**kern` target. This raises input complexity without increasing target entropy.

context-free grammar enforced during inference (Sec. 3.4). The decoder can therefore focus entirely on the visual-semantic alignment task, effectively reducing source-specific label noise to zero before training begins.

3.3.3 Asymmetric Semantic Augmentation

Real scores carry far higher semantic density than standard synthetic renders. Beyond pitch and rhythm, they include dynamics, pedal markings, tempo text, and other performance directives that add clutter and partial occlusion. If trained only on sparse renders, the model can overfit to empty whitespace and default engraving statistics rather than to the underlying notation graph.

To address this without inflating target entropy, we use asymmetric semantic augmentation (Fig. 4). During rendering, we inject musically valid but structurally extraneous `**kern` markers into the source file. These markers are rendered into the image and increase local visual density, but we explicitly omit them from the ground-truth text target. As illustrated in the figure, this visually mimics the dense performance metadata of real scores while preserving a clean, canonical target sequence.

By asking the model to predict the same canonical sequence from a denser image, we decouple visual layout density from target sequence complexity. The encoder must learn invariant representations of the core musical graph and treat performance metadata as distractors instead of vocabulary-bearing symbols.

3.3.4 Domain-Aware Visual Degradation

To close the remaining gap between normalized synthetic targets and real scans, we apply heavy raster-stage degradation during rendering (Fig. 5). Standard notation software produces pages with uniform spacing, fixed fonts, and clean white backgrounds. If trained only on these renders, the model can anchor on synthetic contrast and staff regularity rather than on the notation itself.

We first apply scanner-centric geometric transforms that introduce rotation, translation, perspective, and horizontal squeeze. This breaks perfectly horizontal staff lines and simulates page curvature or camera capture. We then apply document-style artifact augmentation with low-ink streaks, ink bleed, brightness variation, noise, JPEG compression, and bleed-through. Before this final artifact stage, the notation is also composited over sampled paper textures.

These degradations shift local pixel statistics without changing the underlying notation graph. The ConvNeXt-V2 encoder must therefore rely on stable spatial relations between staves, noteheads, stems, beams, and barlines rather than on clean rendering defaults.

3.4. Layered Constrained Decoding Engine

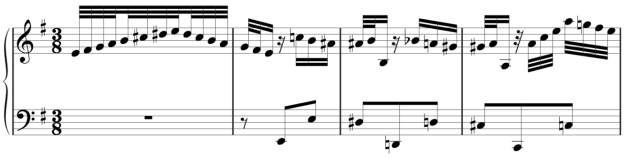
While target canonicalization produces cleaner training targets, downstream applications such as rendering and playback still require formally valid `**kern` sequences. Because `**kern` serializes a dense 2D grid into a 1D text stream, local token validity is insufficient; the output must adhere to global, cross-line dependencies. For this setting, we implement a custom four-layer constrained decoding stack. Because tracking complex 2D state across multiple branching hypotheses is computationally prohibitive, this engine operates strictly via deterministic greedy decoding, trading search-space exploration for structural guarantees.

1. Syntactic Grammar Masking. We compile a standard GBNF grammar against our vocabulary using the `xgrammar` library [21]. This defines a deterministic finite automaton that tracks the local context state s_t induced by the prefix $y_{<t}$. At each step, we compute the admissible token set $\mathcal{A}(s_t) \subseteq \mathcal{V}$ and apply a hard mask to the model logits $\mathbf{z}_t$:

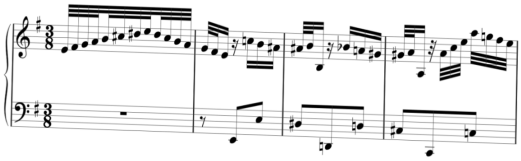
$$z'_t(v) = \begin{cases} z_t(v) & \text{if } v \in \mathcal{A}(s_t) \\ -\infty & \text{otherwise} \end{cases} \tag{1}$$

This layer strictly enforces local structural logic, such as the canonical ordering of properties (pitch, duration, accidentals) within a single note token.

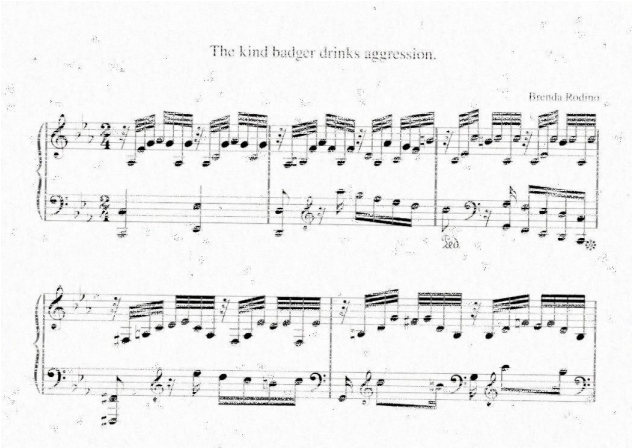
2. Stateful Semantic Rules. To enforce 2D structural invariants that cannot be captured by GBNF, we implement a custom stack of Python-side `LogitsProcessor` modules. These maintain an explicit, delimiter-aware (tabs and newlines) state machine across the generated prefix:



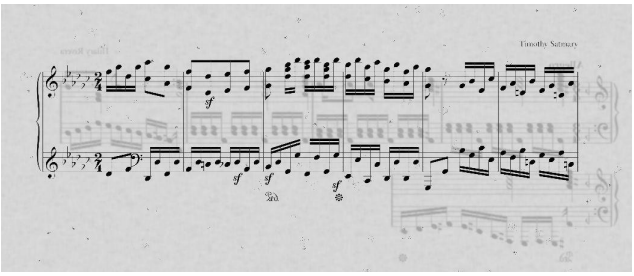
(a) Clean synthetic render



(b) Geometric transformation



(c) Low-ink morphology



(d) Bleed-through and paper artifacts

Figure 5. Domain-aware visual degradation makes the synthetic-to-real gap visible. We transform clean renders (a) with geometric distortions, like spatial squeezing, rotation, and camera perspective shifts (b), low-ink morphology (c), and document artifacts such as bleed-through and textured paper (d). This is designed to discourage the vision encoder from overfitting to synthetic rendering defaults and encourage tolerance to real-world document variations.

- *Spine Integrity:* The processor tracks the active number of parallel voices (spines). It dynamically masks tab and newline tokens to guarantee that every horizontal line emits exactly the correct number of fields, accounting for dynamic inline voice splits (`*^`) and merges (`*v`).
- *Rhythmic Mathematics:* The processor tracks the active time signature and the accumulated duration of the current measure. It actively masks continuation tokens or barlines that would violate the mathematical capacity of the measure, guaranteeing valid temporal synchronization across all parallel spines.

4. Experiments

4.1. Training Details

We train all models using PyTorch Lightning with `bf16` mixed precision and `torch.compile` enabled for accelerated throughput. All main experiments were trained from scratch on a single NVIDIA RTX 5090 GPU in approximately 5 hours. The final dataset comprises 361,938 training examples and 3,951 held-out test examples.

Optimization Strategy. We optimize the network using fused AdamW with an effective batch size of 72, accumulated over 6 micro-batches. To preserve the pretrained ConvNeXt-V2 feature representation, we apply differential learning rates: 3×10^{-4} for the vision encoder and 1×10^{-3} for the randomly initialized projector and Transformer decoder. The learning rate schedule consists of a 500-step linear warmup from 10% of the base rate, followed by a cosine annealing decay to a shared minimum of 5×10^{-5}.

Regularization and Stopping. We apply a weight decay of 0.01 exclusively to weight matrices, leaving biases and normalization parameters unpenalized. To stabilize autoregressive modeling, we enforce a gradient clipping norm of 1.0 and apply a label smoothing factor of 0.1 to the cross-entropy loss. We train for a maximum of 10 epochs, employing early stopping with a patience of 12 validation intervals, evaluated every 1,000 update steps. Checkpointing and early stopping monitor the $\text{CER}_{\text{no_ties_beams}}$ metric on the historical Polish validation split to prevent overfitting to the synthetic training distribution.

Evaluation Distributions. To quantify the simulation-to-reality gap, we evaluate on two domains that serve as target and control environments:

- **Target Domain (Real Scans):** A historical real-world split of 97 Polish scanned scores [22]. It contains print-and-scan degradation, historical engraving artifacts, and paper noise, making it the main test of zero-shot transfer. We removed seven examples from the original 104-score split because their target sequences exceed our maximum context length of 2,048 tokens.
- **Control Domain (Synthetic):** A benchmark of 6,864 cleanly rendered examples generated with Verovio, which we release with this work [23]. This provides a

clean upper bound for in-domain sequence modeling. The performance gap between the Verovio control set and the Polish scanned set directly measures the remaining sim-to-real gap for end-to-end OMR systems.

4.2. Metrics and Decoding

Because raw character-level metrics often fail to capture the true structural fidelity of 2D musical graphs, we evaluate our system using two specialized domain metrics alongside standard character-level metrics [20, 24].

Tree Edit Distance with Note Flattening (TEDn): TEDn evaluates the hierarchical structure of predicted MusicXML files against the ground truth. Standard tree edit distance disproportionately penalizes single-note errors because a single musical note contains many nested XML child nodes [20, 25]. TEDn mitigates this by flattening note sub-trees into compact string representations that encode pitch, duration, and stem direction before computing edit distance. The final score is normalized by the structural cost of the ground-truth tree. We report the standard TEDn, which measures the distance after routing the prediction through standard format converters. Lower scores indicate fewer structural errors [20].

OMR Normalized Edit Distance (OMR-NED): While TEDn relies on XML trees, OMR-NED provides a format-agnostic visual evaluation [24]. It computes the edit distance between constituent musical symbols rather than raw text tokens [24]. Crucially, OMR-NED enforces strict temporal offset matching: notes, rests, and non-note directions are only directly compared if they occur at the exact same temporal position within a measure [24]. Otherwise, they are treated as separate insertions (I) and deletions (D) [24]. The error is defined as $(I + D)/(N_1 + N_2)$, where N_1 and N_2 are the total symbol counts of the respective scores [24]. We report OMR-NED as a normalized error percentage (↓), where lower values indicate better visual-semantic alignment [24].

Additionally, during intermediate validations, we report $\text{CER}_{\text{no_ties_beams}}$, which ignores tie and beam symbols to isolate pitch and rhythm transcription fidelity.

Decoding Configuration. Our primary benchmark relies on the OMR-NED metric, which evaluates visual-semantic alignment directly on the native `**kern` syntax without requiring conversion to intermediary structural formats like MusicXML. We therefore report Transcoda with unconstrained beam search (beam width 3) for all main benchmark results. This setting best reflects the model's learned alignment quality. Our stateful constrained decoding stack (Sec. 3.4) serves a different purpose: it provides strict structural guarantees for downstream rendering or playback engines, but its current greedy adapter trades search depth for validity. Reporting unconstrained beam search for the main empirical baseline lets us separate raw sequence-modeling accuracy from the optional validity layer. We evaluate all comparative models using their published inference configurations, tuning sequence length and beam search where necessary to ensure fair comparison.

4.3. Ablations

To isolate the contributions of our proposed data engine and inference stack, we perform a two-tiered ablation study. Because our pipeline addresses two distinct challenges, namely fundamental sequence modeling and visual domain shift, we evaluate these components on the specific evaluation sets they target.

In-Domain Representation Capacity. Table 3 isolates the components responsible for establishing a stable visual-semantic alignment. We evaluate these on the clean, synthetic Verovio control set to remove the confounding variable of visual domain shift.

- *Target Canonicalization:* Removing deterministic normalization (Sec. 3.3.2) forces the model to learn raw annotator style and redundant markup. This significantly degrades sequence prediction, confirming that reducing target entropy is a prerequisite for stable OMR.
- *Constrained Decoding:* Replacing constrained greedy decoding with unconstrained search improves raw edit-distance metrics on this clean benchmark. This shows that once targets are canonicalized, the constraint stack is not needed to avoid collapse in the control setting. Its role is to guarantee formal validity when that property matters more than raw matching score.

Table 3. In-domain representation ablations evaluated on the synthetic Verovio control set. Parentheses indicate the absolute change (Δ) relative to Full Transcoda. In both ablation tables, Full Transcoda denotes the full training pipeline evaluated with default unconstrained greedy decoding. Additional inference rows apply constrained decoding or beam search on top of this shared baseline. This isolates sequence-modeling capacity from visual domain shift.

Model Configuration	**CER ↓**	**OMR-NED ↓**
Full Transcoda	**3.79**	**18.56**
w/o Target Canonicalization	27 (+23.21)	82.51 (+63.95)
+ Constrained Decoding	3.80 (+0.01)	18.59 (+0.03)
+ Beam Search	2.72 (-1.07)	18.22 (-0.34)

The Verovio control set makes the main causal pattern clear. Removing target canonicalization collapses performance from 18.56 to 82.51 OMR-NED. Adding constrained decoding changes the clean-control result only marginally,

while adding beam search yields the best raw score at 18.22 OMR-NED. This indicates that canonicalization, not the constraint engine, is what stabilizes autoregressive generation on clean data. The constrained decoder remains useful as an optional validity layer for downstream tools.

Out-of-Domain Transfer Robustness. Table 4 evaluates the components intended to improve transfer on the historical Polish scan dataset. The empirical results show that heuristic synthetic augmentations do not solve this extreme transfer regime:

- *Limits of Visual Degradation:* Removing raster-stage visual degradation improves both metrics, lowering OMR-NED by 3.15 points and CER by 4.69 points.
- *Limits of Semantic Augmentation:* Removing asymmetric semantic augmentation improves OMR-NED by 1.11 points, although CER increases by 14.06 points. This means the augmentation does not provide a consistent transfer benefit in this regime.
- *Hypothesis:* We attribute this failure to the extreme length extrapolation in the Polish dataset. Its median target length is 1,372 tokens, compared with 225 tokens during training. At this scale, autoregressive decoding becomes highly sensitive to small visual errors. Synthetic degradations can force the encoder to drop subtle cues that are needed to maintain long-range alignment, so added noise can worsen rather than reduce transfer error.

The Cost of Structural Validity. Table 4 also reveals a clear trade-off in the inference engine. Relative to the default unconstrained greedy baseline, adding constrained decoding guarantees absolute structural validity but degrades raw sequence-matching metrics (CER +6.48, OMR-NED +0.17). By contrast, adding beam search improves raw matching (CER -0.31, OMR-NED -4.07). Because strict grammar enforcement operates greedily, forcing a valid continuation even when the encoder misinterprets a highly degraded region, it can induce larger downstream structural shifts that string-based edit distances heavily penalize.

4.4. Discussion and Limitations

Transcoda improves the zero-shot baseline, but the Polish scan results make clear that real-world historical OMR remains unsolved. The best configuration still reaches 76.03% OMR-NED. This matters for the paper's claim structure: the main positive result is not that Transcoda solves transfer, but that a small model with canonicalized targets outperforms much larger baselines while exposing where synthetic-first training still breaks. The failure mode on the Polish dataset reflects two distribution shifts that our synthetic pipeline does not cover:

Table 4. Out-of-domain transfer ablations evaluated on the historical Polish scan dataset. We evaluate the impact of our proposed data and inference constraints via a leave-one-out methodology. Parentheses indicate the absolute change (Δ) relative to Full Transcoda, which here also denotes the full training pipeline with default unconstrained greedy decoding. Additional inference rows apply constrained decoding or beam search on top of this shared baseline. Beam search used 3 beams.

Model Configuration	**CER** $\downarrow$	**OMR-NED** $\downarrow$
Full Transcoda	**46.5**	**80.1**
Data Engine Ablations		
w/o Asym. Augmentation	60.56 (+14.06)	78.99 (-1.11)
w/o Visual Degradation	41.81 (-4.69)	76.95 (-3.15)
Inference Ablations		
+ Constrained Decoding	52.98 (+6.48)	80.27 (+0.17)
+ Beam Search	46.19 (-0.31)	76.03 (-4.07)

1. **Unstructured Historical Distractors.** Real archives contain visual elements absent from our synthetic pipeline. Library stamps, pencil scribbles, handwritten catalog numbers, and non-musical typography such as inline song lyrics introduce severe occlusions. The vision encoder never sees these specific distractors during training, so they trigger local transcription failures.
2. **Layout Density and Length Extrapolation Shift.** The spatial statistics of our training data and target evaluations differ sharply. The synthetic training set has a median sequence length of 225 tokens. The Polish dataset has a median of 1,372 tokens. This $6\times$ length jump makes autoregressive decoding much less tolerant to local visual mistakes. Our ablations suggest that heuristic visual degradations are not enough in this regime and can even hurt transfer.

The constrained decoder remains useful when a downstream pipeline requires formally valid `**kern`, but it does not fix the underlying alignment problem. Closing the remaining transfer gap will likely require methods that adapt representations to real archives directly, such as unsupervised domain adaptation on unlabelled scans, rather than more synthetic noise injection alone.

5. Conclusion

We demonstrated that in highly structured sequence modeling tasks like OMR, unconstrained target entropy causes autoregressive collapse under domain shift. By canonicalizing training targets, we established Transcoda, a compact baseline that outperforms massive models on real-world historical scans. While optional constrained decoding guarantees formal validity for downstream rendering, stable generation fundamentally relies on controlling

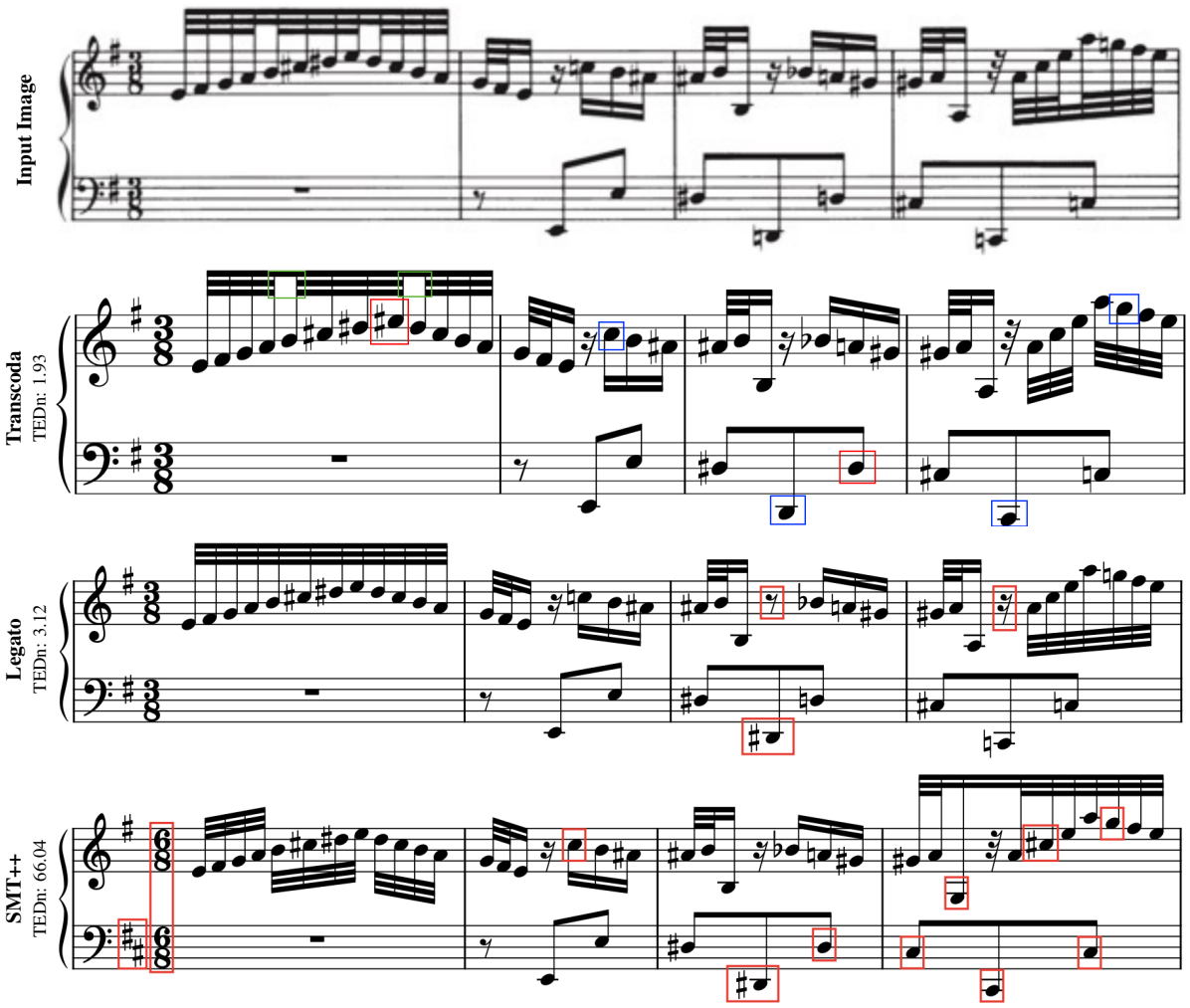


Figure 6. Qualitative comparison on a physical scan of Bach's Duetto No. 1 in E minor (BWV 802). The top row shows the raw input image, followed by the rendered predictions of Transcoda and the baseline models. We annotate specific transcription phenomena in Transcoda's output: **Red boxes** indicate strict structural or pitch errors. **Blue boxes** mark omitted courtesy accidentals; while these represent a visual deviation from the original engraving, they preserve strict semantic equivalence, as the played pitch remains mathematically correct. The **Green box** highlights a subtle gap in the beaming structure that Transcoda successfully recovers. A fine-grained visual detail that the baseline models fail to capture.

the training targets rather than scaling model size. Future work must bridge the remaining simulation-to-reality gap not just through heuristic synthetic noise, but by adapting representations directly to the unstructured layout density and archival distractors of physical documents.

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